

HOTELLING'S T^2 -DISTRIBUTION FOR A MIXTURE OF TWO NORMAL POPULATIONS

D.G. Kabe
and

Bowling Green State University, Bowling Green,
Ohio 43403-0221

A.K. Gupta

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Summary:

The density of Hotelling's T^2 -statistic when sampling from a mixture of two normal populations, under the assumption of the proportionality of the mean vectors, has been derived by Srivastava and Awan (1982), and Srivastava (1983). They do not express the distribution in the standard form. The present paper expresses this distribution in the standard form.

1. INTRODUCTION

Let $Y, p \times n$, be a sample of size n from a mixture of two p -variate normal populations,

$$(1) \quad g(\mathbf{y}) = \lambda g(\mathbf{y}, \mu_1, \Sigma) + (1 - \lambda)g(\mathbf{y}, \mu_2, \Sigma), \quad 0 < \lambda < 1,$$

where for $i = 1, 2$

$$(2) \quad g(\mathbf{y}, \mu_i, \Sigma) = (2\pi)^{-p/2} |\Sigma|^{-\frac{1}{2}} \exp\left\{-\frac{1}{2} \text{tr} \Sigma^{-1} (\mathbf{y} - \mu_i)(\mathbf{y} - \mu_i)'\right\}.$$

Then the density of

$$(3) \quad T^2 = n(n-1) \bar{\mathbf{y}}' S^{-1} \bar{\mathbf{y}},$$

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is desired where $\bar{\mathbf{y}}(p \times 1)$ is the sample mean vector and $S(p \times p)$ is the sample dispersion matrix. Srivastava and Awan (1982, p.95, equation (6.12)), and Srivastava (1983, pp.1490-1496) derived the density of (3) in a rather complicated nonstandard form. This paper expresses this density in a standard form.

Some useful results are stated in the next section and the density of (3) is derived in Section 3.

2. SOME USEFUL RESULTS

Let $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n$ be n p -component vectors and $t_1, \dots, t_n$ be any real numbers, then the identity

$$(4) \quad \sum_{i=1}^n \mathbf{y}_i \mathbf{y}_i' = (t_1 \mathbf{y}_1 + \dots + t_n \mathbf{y}_n)(t_1 \mathbf{y}_1 + \dots + t_n \mathbf{y}_n)' + \sum_{i < j} (t_i \mathbf{y}_j - t_j \mathbf{y}_i)(t_i \mathbf{y}_j - t_j \mathbf{y}_i)',$$

holds when $t_1^2 + \dots + t_n^2 = 1$. Choosing $t_i = n^{-\frac{1}{2}}$, we find that

$$(5) \quad (\mathbf{r} \mathbf{y}_1 \mathbf{y}_1' + (n-r) \mathbf{y}_2 \mathbf{y}_2') - \frac{1}{n} (\mathbf{r} \mathbf{y}_1 + (n-r) \mathbf{y}_2)(\mathbf{r} \mathbf{y}_1 + (n-r) \mathbf{y}_2)' \\ = \frac{r(n-r)}{n} (\mathbf{y}_1 - \mathbf{y}_2)(\mathbf{y}_1 - \mathbf{y}_2)'.$$

The following integral occurs very frequently in noncentral multivariate distribution theory and is also needed in the sequel.

$$(6) \quad \int_{-1}^1 (1-t^2)^{n-\frac{1}{2}} \exp\{\theta t\} dt \\ = \Gamma(n + \frac{1}{2}) \sum_{\alpha=0}^{\infty} \frac{\sqrt{\pi} \theta^{2\alpha}}{\alpha! (n+\alpha)! 2^{2\alpha}} = \sqrt{\pi} \frac{\Gamma(n + \frac{1}{2})}{\Gamma(n+1)} {}_0F_1(n; \frac{1}{4} \theta^2).$$

3. DISTRIBUTION THEORY

Using (5), Srivastava and Awan (1982, pp.83-85, equations (3.2), (3.4)), showed that, for a fixed $\mathbf{r}, \bar{\mathbf{y}}$ and S are independent. The density of $\bar{\mathbf{y}}$ is

$$(7) \quad g(\bar{y}) = K \exp \left\{ -\frac{1}{2} tr \, n \Sigma^{-1} \left[\bar{y} - \left(\frac{r\mu_1 + (n-r)\mu_2}{n} \right) \right] \right. \\ \left. \cdot \left[\bar{y} - \left(\frac{r\mu_1 + (n-r)\mu_2}{n} \right) \right]' \right\}$$

and the density of S is

$$(8) \quad g(S) = k \sum_{r=0}^n \binom{n}{r} \lambda^r (1-\lambda)^{n-r} \exp \left\{ -\frac{1}{2} tr \Sigma^{-1} S \right\} \\ \cdot \exp \left\{ -\frac{1}{2} \frac{r(n-r)}{n} \delta' \Sigma^{-1} \delta \right\} |S|^{(n-p-2)/2} \\ \cdot {}_0F_1 \left(\frac{n-1}{2}; \frac{r(n-r)}{n} \frac{\delta' \Sigma^{-1} S \Sigma^{-1} \delta}{4} \right),$$

where $\delta = \mu_1 - \mu_2$, and K as a generic letter denotes the normalizing constants of density functions in this paper. By using a generalized version of (5), Tan ((1978), p.51, equation (3.5)) generalizes (8) to a mixture of $k < n$ normal populations.

Now the density of Hotelling's T^2 is the density of the first element of a noncentral multivariate beta matrix. The density of such a matrix for $k = 2$ is derived by Kabe (1963), and by Sitgreaves (1952). We derive the density of T^2 for $k = 2$ by following Kabe (1963).

Assuming μ_1 and μ_2 to be proportional, we set

$$(9) \quad \sqrt{n} \bar{y} = y, \quad (r\mu_1 + (n-r)\mu_2) = \sqrt{n} \theta_1 \xi, \\ (r(n-r)/n)^{\frac{1}{2}} \delta_2 = \theta_2 \xi,$$

$$(10) \quad G = S + yy' = \Sigma^{\frac{1}{2}} Q B Q' \Sigma^{\frac{1}{2}}, \quad y = \Sigma^{\frac{1}{2}} Q x, \\ \xi' \Sigma^{-1/2} Q = (\xi' \Sigma^{-1} \xi)^{\frac{1}{2}} (1, 0, \dots, 0),$$

where $Q, p \times p$, is an orthogonal matrix. Note that

$$(11) \quad m = x' B^{-1} x \\ = (y' S^{-1} y) / (1 + y' S^{-1} y) = [T^2 / (n-1) / (1 + T^2 / (n-1))].$$

Now using (7), (8), (9), (10), the joint density of B and $\mathbf{x}$ is

$$(12) \quad g(B, \mathbf{x}) = K \phi(r, \lambda, \theta_1, \theta_2, \xi) \exp \left\{ -\frac{1}{2} \text{tr} B \right\} |B - \mathbf{x} \mathbf{x}'|^{(n-p-2)/2} \\ \cdot \exp \{ \theta_1 (\xi' \Sigma^{-1} \xi)^{\frac{1}{2}} (1, 0, \dots, 0) \mathbf{x} \} \\ \cdot {}_0F_1 \left(\frac{1}{2}(n-1); \frac{1}{4} \theta_2^2 (\xi' \Sigma^{-1} \xi) (b_{11} - x_1^2) \right),$$

where x_1 is the first component of the vector $\mathbf{x}$ and

$$(13) \quad \phi(r, \lambda, \theta_1, \theta_2, \xi) = \sum_{r=0}^n \lambda^r (1-\lambda)^{n-r} \binom{n}{r} \exp \left\{ -\frac{1}{2} \xi' \Sigma^{-1} \xi (\theta_1^2 + \theta_2^2) \right\}.$$

If $\mathbf{x}$ is a p component vector, $B = (b_{ij})$ a $p \times p$ symmetric positive definite matrix, and D a $q \times p$ given matrix of rank $q < p$, then Kabe (1965) has shown that

$$(14) \quad \int_{\mathbf{x}' B^{-1} \mathbf{x} = m, D \mathbf{x} = \mathbf{v}} f(\mathbf{x}' B^{-1} \mathbf{x}, D \mathbf{x}) d\mathbf{x} = \frac{\pi^{(p-q)/2}}{\Gamma(\frac{1}{2}(p-q))} |B|^{\frac{1}{2}} |D B D'|^{-\frac{1}{2}} f(M, \\ \cdot (m - \mathbf{v}' (D B D')^{-1} \mathbf{v})^{(p-q-2)/2}.$$

Now using (14) we evaluate the integral

$$(15) \quad \int (1 - \mathbf{x}' B^{-1} \mathbf{x})^{(n-p-2)/2} \exp \{ \theta_1 (\xi' \Sigma^{-1} \xi)^{\frac{1}{2}} (1, 0, \dots, 0) \mathbf{x} \} d\mathbf{x},$$

over the range

$$(16) \quad \mathbf{x}' B^{-1} \mathbf{x} = M, \quad (1, 0, 0, \dots, 0) \mathbf{x} = x_1,$$

and find the value of the integral to be

$$(17) \quad K |B|^{\frac{1}{2}} (1-m)^{(n-p-2)/2} \left(m - \frac{x_1^2}{b_{11}} \right)^{(p-3)/2} (b_{11})^{-\frac{1}{2}} \\ \exp \{ \theta_1 (\xi' \Sigma^{-1} \xi)^{\frac{1}{2}} x_1 \}.$$

We next set $x_1 = b_{11}^{\frac{1}{2}}z$, and using (12) find the joint density of B, m , and z to be

(18)

$$g(B, m, z) = K \phi(r, \lambda, \theta_1, \theta_2, \xi) \exp \left\{ -\frac{1}{2} \text{tr} B \right\} |B|^{(n-p-1)/2} \\ \cdot \exp \{ \theta_1 (\xi' \Sigma^{-1} \xi)^{\frac{1}{2}} b_{11}^{\frac{1}{2}} z \} (1-m)^{(n-p-2)/2} (m-z^2)^{(p-3)/2} \\ \cdot {}_0F_1 \left(\frac{1}{2}(n-1); \frac{1}{4} \theta_2^2 (\xi' \Sigma^{-1} \xi) b_{11} [(1-m) + (m-z^2)] \right).$$

It may now be noted that

(19)

$${}_0F_1 \left(\frac{1}{2}(n-1); \frac{1}{4} \theta_2^2 (\xi' \Sigma^{-1} \xi) b_{11} [(1-m) + (m-z^2)] \right) \\ \sum_{\alpha_1=0}^{\infty} \sum_{\alpha_2=0}^{\infty} \\ \frac{\Gamma \left(\frac{n-1}{2} + \alpha_1 + \alpha_2 + 1 \right) \theta_2^{2(\alpha_1+\alpha_2)} (\xi \Sigma^{-1} \xi)^{\alpha_1+\alpha_2} b_{11}^{\alpha_1+\alpha_2} (1-m)^{\alpha_1} (m-z^2)^{\alpha_2}}{2^{2(\alpha_1+\alpha_2)} \alpha_1! \alpha_2!}$$

From (18) we first integrate out all the elements of B , except b_{11} , we then integrate (18) with respect to z by using (6), and finally integrate out b_{11} . The result is the density of $m = [T^2/(n-1)]/(1+T^2/(n-1))$ which is

(20)

$$g(m) = K \sum_{r=0}^n \binom{n}{r} \lambda^r (1-\lambda)^{n-r} \exp \left\{ -\frac{1}{2} (\theta_1^2 + \theta_2^2) \xi' \Sigma^{-1} \xi \right\} m^{(p-2)/2} \\ (1-m)^{(n-p-2)/2} \sum_{\alpha_1=0}^{\infty} \sum_{\alpha_2=0}^{\infty} \sum_{\beta=0}^{\infty} \frac{(1-m)^{\alpha_1} m^{\alpha_2+\beta} \theta_2^{2(\alpha_1+\alpha_2)} \theta_1^{2\beta}}{2^{\alpha_1+\alpha_2+\beta} \alpha_1! \alpha_2! \beta!} \\ \frac{\Gamma \left(\frac{n+1}{2} + \alpha_1 + \alpha_2 \right) \Gamma \left(\frac{p-1}{2} + \alpha_2 \right) (\xi \Sigma^{-1} \xi)^{\alpha_1+\alpha_2+\beta}}{\Gamma \left(\frac{1}{2} p + \alpha_2 + \beta \right)}.$$

From (20) the density of T^2 is easily obtained.

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